Product Requirement Document (PRD)

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1. Introduction

1.1 Purpose

This ERP system is designed to digitize and simplify the supply chain operations for a uPVC profile trading business. It will connect traders, distributors, and fabricators into a single workflow.

The ERP will handle: - Invoicing (GST & Non-GST). - Quotation & Production (with cutting optimization). - Inventory management. - SaaS monetization (fabricators pay per square foot for production plans). - Dynamic pricing per region/customer type. - AI-powered OCR for handwritten measurement sheets.

It will be mobile-friendly, easy-to-use, and icon-driven, keeping in mind low literacy users.

1.2 Objectives

- Simplify order, production, and quotation processes.
- Enable real-time linkage between trader → distributor → fabricator.
- Provide production optimization (1D & 2D cutting plans, BOM).
- Support GST and Non-GST invoices.
- Generate reports for CA.
- Introduce a scalable SaaS revenue model for production plan usage.

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2. Stakeholders

- **Trader (Admin):** Manage supply, invoices, stock, pricing rules, and reports.
- **Distributor:** Accept invoices, manage sales to fabricators, stock, and quotations.
- **Fabricator:** Input measurements, generate production plan & quotations, pay per sft usage.
- **ERP Provider (System Owner):** Manages SaaS billing, subscriptions, and revenue.
- **CA/Accountant (Indirect User):** Uses reports for taxation and accounting.
- **Company Teams:** Each company will have its own team, and individual admins within the company can define roles and permissions for their team members based on their authority.

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3. Workflow

3.1 Supply Chain Workflow

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Trader → Distributor → Fabricator → Client
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3.2 Quotation & Production Workflow

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Input → Validation → OCR (if image) → Optimizer → Output
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3.3 SaaS Billing Workflow

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Fabricator Usage → SFT Calculation → Deduction/Invoice → Payment
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4. Key Features (Phase 1 Scope)

4.1 User Management

- Role-based login (Trader, Distributor, Fabricator).
- OTP/email login.
- Multi-language support (icons + minimal text).
- Company admins can define roles and permissions for their team members.

4.2 Invoicing

- Linked invoicing chain (Trader → Distributor → Fabricator).
- GST invoice (with tax breakdown).
- Non-GST invoice option.
- Payment tracking (pending/paid/overdue).

4.3 Quotation Module

- Inputs (image/PDF/form/drawing).
- OCR-powered auto-reading of handwritten inputs.
- Auto-generation of quotations with:
    - Detailed 2D drawings.
    - BOM.
    - Region/customer-specific pricing.
    - GST or Non-GST toggle.

4.4 Production Module

- Optimized 1D & 2D cutting plans.
- BOM summary (profiles, glass, hardware, strips).
- Multi-project batch processing.
- SaaS billing integration (pay per sft).

4.5 Inventory

- Stock tracking (profiles, glass, hardware).
- Auto-deduction after invoice or production plan approval.
- Distributor/fabricator stock visibility.
- Works for non-uPVC products.

4.6 Reports

- Sales & purchase reports (period-wise).
- GST summary (for GST invoices).
- Non-GST sales reports.
- Export to Excel/PDF (for CA).

4.7 SaaS Billing

- Dynamic pricing rules per region/customer.
- Pay-per-sft billing for production plans.
- Fabricator wallet (credits).
- Online payments.
- Admin revenue reports.

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5. Non-Functional Requirements

- Mobile-first, responsive design.
- Works on low-end Android devices.
- Data encryption & role-based access.
- Scalable SaaS model.
- Multi-language UI.
- Offline-first capability.

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6. Success Metrics

- 90% of users use ERP without training.
- Production errors reduced by 80%.
- Quotation & invoicing cycle 50% faster.
- SaaS adoption rate >70% fabricators.
- Measurable recurring SaaS revenue growth.

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7. Phase 1 Deliverables

- Web ERP application.
- Mobile-optimized UI (Android/iOS).
- Trader/Distributor/Fabricator dashboards.
- Invoicing (GST & Non-GST).
- Quotation & Production module (with OCR + cutting optimization).
- Inventory management.
- Reports (Excel/PDF export).
- SaaS billing (per sft usage).
- Admin dashboard (pricing, billing, revenue).
- Company admin controls for role and permission management within teams.

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8. Future Enhancements (Phase 2)

- Voice-to-text measurement input.
- WhatsApp/Telegram order notifications.
- API integrations with Tally/Zoho.
- Dynamic pricing automation with AI.
- Dealer/fabricator e-commerce ordering portal.